

xFC Multi-Ethernet Platform Quick Start Guide PE

Document Index: R08

2016-04-12

fw, DC-IA/EDH4

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1 Document Information

1.1 Overview

Project	---
Project Manager	---
Project Number	---
Title	xFC Multi-Ethernet Platform Quick Start Guide PE
Version	R08
Created	2015-10-13
Author	fw
Department	DC-IA/EDH4
Last saved at	2016-04-12 09:19
File Name	Fw_BG_YAD20_MEP-XFC-QuickStartGuidePE.docx

Table 1: Overview

1.2 Document History

Document Index	Date (CheckIn)	Created / Changed	Verified	Confirmed	Description
R01	2015-07-18	fw			First Edition
	2015-07-20	fw			A subnet check will no longer be performed when writing IP address, subnet mask and gateway address.
	2015-07-27	rs			Include counter measures for specific diagnosis.
	2015-07-28	fw			Added note about IP address usage with PNIO.
R02	2015-07-31	fw			Added relevant host diagnoses
R03	2015-08-03	fw			Modified Network Status LED description Modified Module Status LED description
R04	2015-08-12	fw			- New section about MEP FWA update - Correction: Invalid or unknown FCs at PDC lead to error FPC- instead of error Fin- - Error FnC-: Possible cause may be also non-corresponding IP address and Gateway address - FnC- countermeasures corrected.
R05	2015-08-14	fw			Added new section about parameter set.
	2015-09-08	fw			Inconsistent IP suite leads to Fin- instead to FnC.
R06	2015-10-13	te			Added Sercos III specific documentation Valid since MEP FWA 01T01.02 and FWA 01V02
	2015-11-09	fw			Introduced H0.40 (alignment parameter)
	2015-12-23	rs			EtherCAT CoE support for CiA402 „Velocity Mode” – w/o Watch Parameters, yet.
	2016-01-13	fw			Changed Module Status LED description (switched description text for red steady and red blinking)
	2016-02-11	fw			Modbus/TCP support introduced
	2016-02-26	tf			Paramaterization of velocity mode for Sercos III
	2016-02-26	rs			Adapted EtherCAT mode selection. Added guide to process data configuration (EtherCAT)

Document Index	Date (CheckIn)	Created / Changed	Verified	Confirmed	Description
R07	2016-03-04	fw			Minor changes and formattings for R07. LED diagnostics for EtherCAT inserted.
	2016-03-04	rs			Operation mode selection required to switch EtherCAT to cyclic communication.
	2016-04-11	rs			Table of user parameters updated in Chapter 5
R08	2016-04-12	fw			Changed hint about engineering connection while updating MEP firmware. Inserted windows TFTP client command line. Introduced new section "Fallacies and Pitfalls".

Table 2: Document History

1.3 Referenced Documents

No.	Document Title and/or Document Type	File Name (Hyperlink included)

Table 3: Referenced Documents

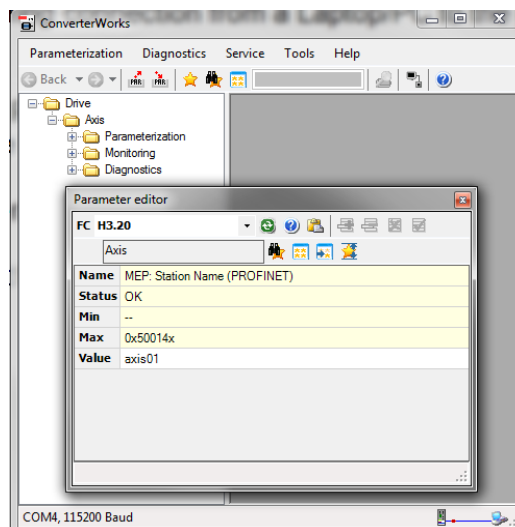
2 Introduction

This document contains a Quick Start Guide for using the Rexroth EFC frequency converter's Multi-Ethernet Platform (MEP) by Parameter Editor (PE). This guide focuses on usage of Industrial-Ethernet derivatives PROFINET, EtherNet/IP, Sercos III, EtherCAT (CoE/EoE) and Modbus/TCP. The guide will be replaced by IndraWorks user dialogs and customer documentation later.

For using the MEP, an engineering connection from a Laptop/PC to the EFC is necessary. Such a connection can be built up...

- ...via Ethernet (Sercos/IP engineering protocol) using IndraWorks. In this case, the MEP can be browsed and the IP address of MEP can be set.
- ...via USB (SIS engineering protocol) using ConverterWorks. Plug the cable and connect.

For simplicity, we are limiting at this guide on the usage of ConverterWorks and its parameter editor:



3 Quick Start

3.1 General Settings

In order to use the MEP for controlling the drive via a fieldbus master, the EFC has to be parametrized as follows:

- Communication Protocol [E8.00]: Set to value 1 for “Option Card”
- First frequency setting source [E0.00]: Set to value 20 for “Communication”
- First run command source [E0.01]: Set to value 2 for “Communication”

3.2 Selecting the Fieldbus

- Select **PROFINET** fieldbus by writing string “PN” to [H3.40].
- Select **EtherNet/IP** fieldbus by writing string “EI” to [H3.40].
- Select **Sercos III** fieldbus by writing string “S3” to [H3.40].
- Select **EtherCAT (CoE)** fieldbus by writing “EC” to [H3.40].
- Select **Modbus/TCP** fieldbus by writing “MB” to [H3.40].

3.3 Parametrizing Fieldbus Addressing

- For **PROFINET**, set the PNIO Device Name by writing [H3.20]. If not assigned by the PNIO controller, manually set IP address [H3.03], Subnet [H3.04], Gateway [H3.05].
- For **EtherNet/IP**, set IP address [H3.03], Subnet [H3.04], Gateway [H3.05].
- For **Sercos III**, the IP address is configured via [S-0-1020], [S-0-1021], [S-0-1022] and [S-0-1039]. These settings will become active after executing [S-0-1048]. A valid IP suite is not required for running Sercos III due to communication is based on broadcasts frames.
- For **EtherCAT**, the IP address configuration is done on master’s (TwinCAT/PLC) side. From EtherCAT state PreOP, Ethernet-over-EtherCAT is started and IndraWorks can be used.
- For **Modbus/TCP**, set IP address [H3.03], Subnet [H3.04], Gateway [H3.05]. A Modbus/TCP client can connect to default TCP port 502. Additionally, a user can specify another port by writing a port number to [H3.51]. However, only *one* client connection is accepted by MEP.

Remark: With PROFINET, it is recommended, either to use a static IP address at fieldbus project, which was already parameterized to the MEP or to ensure, that the IP address assigned dynamically by the PNIO controller at fieldbus startup is equal to the parameterized IP address at MEP.

3.4 Parametrizing Process Data Configuration

3.4.1 PROFINET and EtherNet/IP

Firstly, create a fieldbus project for the specific fieldbus master. The GSDML file needed for PROFINET and EDS file needed for EtherNet/IP are available, respectively. Setting up fieldbus projects for EFC MEP is equivalent to known IndraDrive MEP: Process data configuration is defined only by process data length. However, the fieldbus-specific flow for creating a master project (e.g. using SIEMENS Step 7 for PROFINET) is not described in this document.

For setting up Process Data Configuration at EFC, write the appropriate list of Function Codes to [H3.30] and [H3.31], respectively. The default process data configuration is as follows:

- Input Data List [H3.30]: Status Word [H0.01], Setting Frequency [d0.02]
- Output Data List [H3.31]: Control Word [H0.00], Frequency Command [H0.10]

Using the parameter editor, the [H3.30] and [H3.31] has to be filled with *numerical representations* of Function Codes [XX.YY]. Such a numerical representation is a two-byte value, where the low-byte is the hex value of YY and the high-byte can be derived from XX using the following table.

Function Code Class	Numerical Representation (High-Byte)
b0...b9	0x00...0x09
d0...d9	0x10...0x19
C0...C9	0x20...0x29
E0...E9	0x30...0x39
U0...U9	0x40...0x49
F0...F9	0x50...0x59
H0...H9	0x60...0x69

Example: The numerical representation of Function Code [H3.30] is 0x631E.

Remarks: The set of allowed Functions Codes for input and output process data is contained at [b8.61] and [b8.62], respectively. Maximum supported input and output process data length is 30 bytes, each. Hence, as current supported process data Function Codes are all two bytes in data length, maximum number of configurable Function Codes is 15. If required at PLC, also a 32-bit alignment in process data can be effectuated by inserting parameter [H0.40] into process data configuration. [H0.40] can be inserted multiple times and represents a 2 byte dummy process data, which has no function behind and which is always zero.

For usage of the Function Codes at Process Data Configuration, please refer to the EFC's User Manual.

3.4.2 Sercos III

Firstly, create a fieldbus project for the specific fieldbus master. The SDDML and SPDML files needed for Sercos III are available.

For setting up Process Data Configuration at EFC, configure a consumer and a producer connection (S-0-1050.x.6) at the fieldbus master. The default process data configuration is as follows:

- **Producer Data List [S-0-1050.0.6]:**
Drive Status [S-0-0135], Status Word [P-0-1098.0.1], Setting Frequency [P-0-1058.0.2]
- **Producer Data List [S-0-1050.1.6]:**
Drive Control [S-0-0134], Control Word [P-0-1098], Frequency Command [P-0-1098.0.10]

Additionally to the default configuration, the mandatory Sercos III configuration is also possible. For a velocity axis (FSP_Drive / operation mode “velocity control”), the process data configuration is as follows:

- **Producer Data List [S-0-1050.0.6]:**
Drive Status [S-0-0135], Velocity Feedback [S-0-0040]
- **Producer Data List [S-0-1050.1.6]:**
Drive Control [S-0-0134], Velocity Command [S-0-0036]

To use velocity mode, the primary operation mode [S-0-0032] needs to be set to value 2. Data scaling is currently limited to default scaling of 10^{-4} rpm.

Remark:

- EFC Function Codes can also be addressed by P-EIDNs. The general mapping from P-EIDNs to FCs is done by a simple mapping function, where an FC is embedded into the IDN and SE part of a specific EIDN:

31 ... 24	23 ... 16	15	14 ... 12	11 ... 0
SI	SE	P/S	SET	IDN
31 ... 24	23 ... 16	15	14 ... 12	11 ... 0
0	Y	1	0	X

Let H.L be the High- and the Lowbyte, respectively, of the numerical representation of a Function Code. Then $EIDN = P-0-x.0.y$ with $x = 1050 + (H \gg 1)$ and $y = 100 \cdot (H \& 0x01) + L$. Let $P-0-x.0.y$ be an EIDN out of the mapped area. Then $FC = H.L$ with $H = ((x - 1050) \ll 1) + ((y \geq 100) ? 1 : 0)$ and $L = y - ((y \geq 100) ? 100 : 0)$.

3.4.3 EtherCAT

An EtherCAT master requires both, an EtherCAT Slave Information (ESI) and an Electronic Datasheet (EDS) file in order to fully support an EtherCAT slave running CoE (CAN over EtherCAT). The former provides a slave device description for EtherCAT PLCs and some information for configuring the EtherCAT communication. The latter describes accessible CAN objects of the device. After establishing communication, still in EtherCAT state PreOp, process data configuration can be carried out by TwinCAT/PLC through CoE.

- Mode selection by CAN index [0x6060]: Either Rexroth Mode (-128) or CiA402 Velocity Mode (2). Resulting Process Data Configurations are as follows:
 - o Rexroth Mode: Consumer {[H0.00], [H0.10]} Producer {[H0.01], [D0.02]}
 - o Velocity Mode: Consumer {[0x6040], [0x6042]} Producer {[0x6041], [0x6044]}
- User parameters can be configured by writing CAN indexes in the CAN format 0x<yyyy>01<xx> (y: CAN index | x: bit length of parameter normally (hex) 10 or 16 bits) to the following lists:
 - o Producer Data List [0x1A15]
 - o Consumer Data List [0x1615]

Currently, following parameters can be configured for process data:

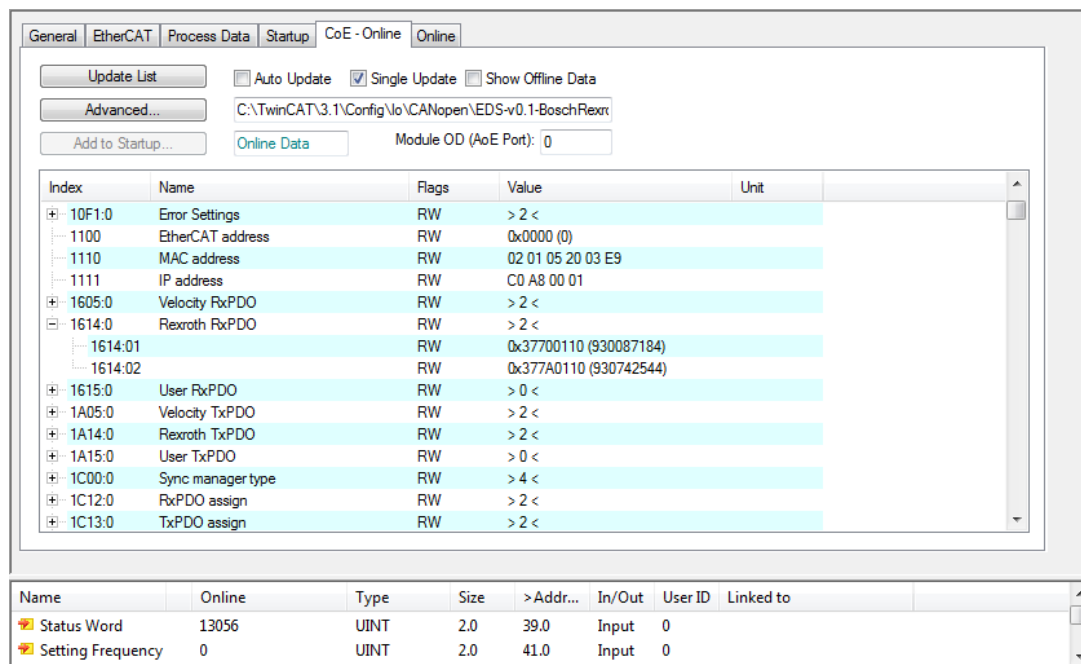
- Producer Data or OUT [0x1615]:
{[H0.00], [H0.10]}
- Consumer Data or IN [0x1A15]:
{[H0.01], [D0.00], [D0.02], [D0.03], [D0.04], [D0.05], [D0.10],
[D0.11], [D0.12], [D0.13], [D0.16], [D0.17], [D0.20], [D0.21]}

Remarks: A mode selection has to be carried out before cyclic data exchange can start. Failing to do so will make MEP deliver “invalid settings” when switching from PreOp to SafeOp. The user parameters can be configured freely. After changing a process data configuration, the first switch to SafeOp yields “unknown” error. A second attempt should succeed and no errors are thrown in case the process data configuration did not change.

3.4.3.1 CAN Indexes for EFC Function Codes

For a corresponding EFC Function Code, the subindex of the data element and the length of the EFC Function Code are required. The subindex of the data is 1 and the length of process data parameters is currently 2 bytes or (16-bit) for EFC. The result of the index calculation is a 16-bit value – it is put together in the following way then: Highest 16-bit are the calculated CAN index, bits 15 to 8 are 1 and the last 8 bits are 16 or 10 in hexadecimal.

In the example of the following figure you can see, that the first element is 0x37700110. 0x3770 is the CAN index calculated from EFC Function Code [H0.00], 0x01 is the subindex and 0x10 is the 16-bit length. Green and red do not change currently. To fully understand the calculation of a CAN index from an EFC Function Code, check the explanation after the picture.



CAN indexes have to be input as the 16 most significant bits, bits 15 to 8 are the subindex that is always 1 and the least 8 significant bits are the length of the parameter in bits, usually 16. To find out, which CAN index corresponds to which EFC Function Code, there is a special conversion function. The conversion function is described as follows:

Let H.L be the High- and Low byte, respectively of the numerical representation of an EFC Function Code, where H is the simple decimal interpretation of the hexadecimal encoding of the Function Class. Example: FC class “d” is encoded by 0x10. The simple decimal interpretation of “0x10” then is “10”. (Hint: This simple trick closes the unused gap in FC class encoding between 0x0A and 0x0F in order to get all FCs mapped into CANopen index manufacturer parameter range between 0x2000 and 0x5FFF.)

Then, the index of the corresponding “Manufacturer Specific Object” is: $I = 0x2000 + H \times 100 + L$.

EFC Function Code Yx.z, where $Y \in \{b,d,C,E,U,F,H\}$, $x \in \{0\dots9\}$, $z \in \{0\dots99\}$

That means:

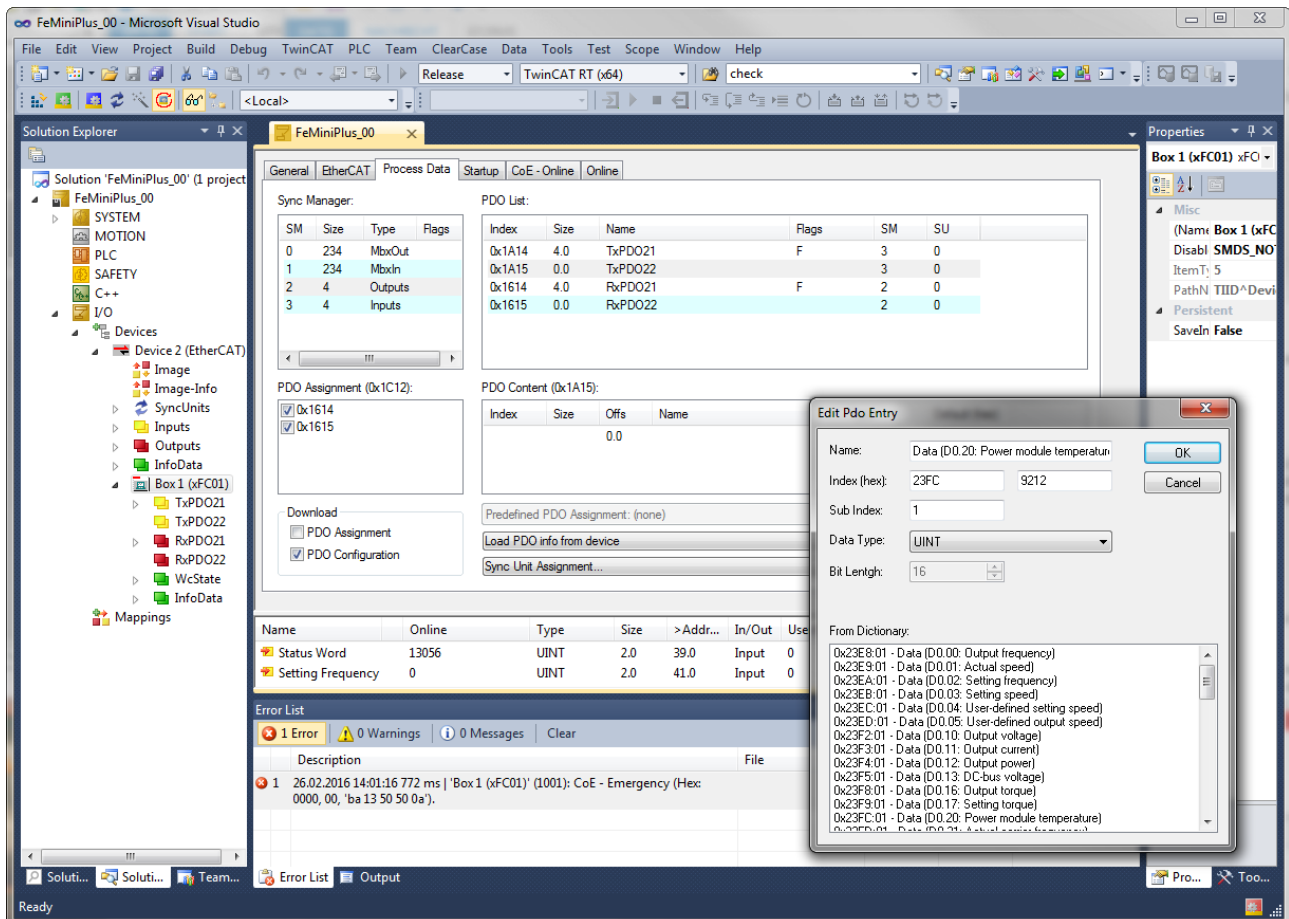
Function Code H.L range (DEC) FC Index (DEC) CAN Index (HEX)

$bx.z \rightarrow \{00\dots09\} \cdot \{0\dots99\} \rightarrow \{0000\dots0999\} \rightarrow \{0x2000\dots0x23E7\}$
 $dx.z \rightarrow \{10\dots19\} \cdot \{0\dots99\} \rightarrow \{1000\dots1999\} \rightarrow \{0x23E8\dots0x27CF\}$
 $Cx.z \rightarrow \{20\dots29\} \cdot \{0\dots99\} \rightarrow \{2000\dots2999\} \rightarrow \{0x27D0\dots0x2BB7\}$
 $Ex.z \rightarrow \{30\dots39\} \cdot \{0\dots99\} \rightarrow \{3000\dots3999\} \rightarrow \{0x2BB8\dots0x2F9F\}$
 $Ux.z \rightarrow \{40\dots49\} \cdot \{0\dots99\} \rightarrow \{4000\dots4999\} \rightarrow \{0x2FA0\dots0x3387\}$
 $Fx.z \rightarrow \{50\dots59\} \cdot \{0\dots99\} \rightarrow \{5000\dots5999\} \rightarrow \{0x3388\dots0x376F\}$
 $Hx.z \rightarrow \{60\dots69\} \cdot \{0\dots99\} \rightarrow \{6000\dots6999\} \rightarrow \{0x3770\dots0x3B57\}$

3.4.3.2 Using TwinCAT3

On tab “Process Data” under dialog element “PDO Content” – relatively to selected index at dialog element “PDO List” – a new process data can be inserted by clicking with the right mouse button and choosing “Insert”. A pre-selection list will open.

Example: For a selection at [0x1A15], selecting [D0.20] as first user process data would look like in the following picture.



After changing the process data configuration, select the menu TwinCAT → Activate Configuration or use the corresponding quick icon. After TwinCAT has gone into Run Mode, switch it back to Config Mode and activate the Free Run mode to switch the PLC and the drive to OP state. If successful, EtherCAT communication then is active.

3.4.4 Modbus/TCP

With Modbus/TCP, setting up Process Data Configuration at EFC is almost same procedure as with PROFINET (see above): Write the appropriate list of Function Codes to [H3.30] and [H3.31], respectively.

The following Modbus/TCP transactions are supported by MEP:

Modbus Function Code	Transaction Name	Max. value of N
3	Read N register words	16
6	Write one register word	-
16	Write N register words	16
23	Read/write N register words	16/16
43 (Subcode 14)	Read Device Identification	-

Besides accessing parameters by their EFC Function Code, there are some special register addresses, that can be used, e.g. for reading/writing the complete process data image. The following table gives an overview:

Register Address	Contents
0x7F00	Control word [H0.00]
0x7F01	Frequency command value [H0.10]
0x7FA0	Status word [H0.01]
0x7FE0	Input Process Data Image as specified by [H3.30]
0x7FF0	Output Process Data Image as specified by [H3.31]

Remark:

- When a Modbus/TCP client established a new connection to the MEP, output process data status initially is set invalid at the MEP. The output data status changes to valid, as soon as all parameters at output process data list are written at least once. The output data status then remains valid, until the TCP connection is closed or terminated.
- Special Register Addresses mentioned above, only can be used without any offset.
Example: It is *not* allowed to use address 0x7FF2 for accessing second output process data item.

3.5 Getting Into Cyclic Data Exchange

When initial fieldbus parametrization from previous subsections was done, reboot the EFC once in order to get the settings active and plug the Ethernet cable between fieldbus master and MEP (ethernet port 1 or port 2).

If all settings are correct, the MEP will get ready for being connected by the fieldbus master. With PROFINET, EtherNet/IP and Modbus/TCP, this is indicated by a green-blinking Network Status LED (H11 for option slot 1 and H21 for option slot 2). Furthermore, if the drive application has no errors pending, the Module Status LED (H12 for option slot 1 and H22 for option slot 2) will also become green-blinking.

Remarks:

- With PROFINET, EtherNet/IP and Modbus/TCP, after a reboot, the Network Status LED remains dark, as long as no ethernet link was established. As soon as the cyclic fieldbus connection was established, the Network Status LED gets steady green.
- With Sercos III, the Sercos LED (Network Status LED) remains dark until CP0 is reached.

However, if cyclic data exchange does not get up, there are some problems about parametrization or fieldbus connection, some diagnostic information can be derived from:

- LED Display Panel Message
- Error Code (Last error type [E9.05])
- Network Status LED / Sercos LED (Hx1)
- Module Status LED (Hx2)
- PHY Status LEDs (Port 1 Hx3 / Port 2 Hx4, Link/Activity, each)

For more details, see subsection 3.6.

3.6 Diagnostics

The following table shows **display messages and error codes**, when using the MEP.

Display Message	Error Code at [E9.05]	Description	Cause	Countermeasures
Fdi (warning) Fdi- (error)	908 or 53	No valid master data (Warning, if drive in STOP mode / Error, if drive in RUN mode)	Cyclic communication had been established, but was stopped because of an error.	1. Check fieldbus master status. 2. Check Ethernet cable and switches. 3. EMC problems at Ethernet environment: Check shielding and laying of cables. 4. Bus load is too high: Reduce Ethernet traffic, built up separate network for fieldbus communication. At the PLC, set application status and/or process data status valid.
			Cyclic communication had been established, but timed out on fieldbus (telegram losses).	
			Cyclic communication is running, but fieldbus master has set data status invalid. (E.g. EtherNet/IP IDLE-Flag set or PROFINET APDU status STOP or with ModbusTCP: output data not written, yet)	
Fin-	904	Initialization failed	Parametrization of MEP has errors. MEP could not start up completely.	Check list of invalid parameters [b7.93] and/or [H3.62] and rewrite invalid parameters with valid values.
			IP address [H3.03] and gateway address [H3.05] are not matching.	Write a consistent set of IP address [H3.03], subnet mask [H3.04] and gateway address [H3.05]. If no gateway is needed, set [H3.05] to 0.0.0.0.
FPC-Occ	902 or 52	Process Data Configuration Invalid / Wrong Process Data Length	Parametrized process data configuration between MEP and fieldbus master are differing. Calculated data lengths [H3.32]/[H3.33] based on the parameters in [H3.30]/[H3.31] differ from [H3.28]/[H3.29] configured at PLC.	Correct process data configuration either at MEP ([H3.30]/[H3.31]) or at PLC.
			Parametrized process data configuration at MEP is invalid. Either there is at least one unknown/unsupported parameter at process data configuration or calculated data lengths [H3.32]/[H3.33] exceed MEP's maximum allowed.	Check process data configuration at [H3.30] and/or [H3.31] for unknown/unsupported parameters and correct process data configuration at MEP ([H3.30]/[H3.31]).
FCE-	906	Internal error ("Critical Error")	Fatal Error or Exception.	Please reboot the EFC, then take a complete parameter set extraction of the EFC and send it to development department.
FnF-	907	Subsystem Corrupted ("No Firmware")	Firmware file corrupted.	Please run a MEP firmware update. If problem persists, exchange the MEP option hardware.
FCd-	901	Watchdog Cyclic Communication	Cyclic Communication timed out.	If the problem persists, [H3.38] could be increased to a number higher than 4.
FnC-	905	Network setup error	Parametrized IP address already present in network. (Network Status LED is steady red.)	Change [H3.03] IP address to a valid IP address in your subnet.
			No DHCP response from DHCP server.	Check if the DHCP server is up and running.
			Fieldbus parametrization at MEP erroneous.	Check your fieldbus project for using a device description file suitable to EFC.
Ocd-	51	Option card error	MEP was successfully detected on device startup, but the internal communication was disturbed.	Try to clear the error by pressing the STOP button. If the error is clearable, then a transient disturbance of internal communication occurred: 1. MEP system overload. Probably bus load is too high: Reduce Ethernet traffic, built up separate network for fieldbus communication. 2. EMC problems: Check environment (e.g. shielding and laying of cables). If the error is not clearable, please reboot the EFC, then take a complete parameter set extraction of the EFC and send it to development department. Furthermore, the problem could be caused either transient or persisting by contacting problems: Check, that option is mounted correctly on base device.

Note, that [E9.05] contains the code of last occurred error. Besides display messaging, only bit 7 (fault) of status word [H0.01] denotes, whether the error is still pending.

With **Sercos III**, the error message and error code are furthermore accessible via [S-0-0095] and [S-0-0390]. Sercos-Communication-related errors are only signaled via these S-parameters and not via [E9.05]!

With **Modbus/TCP**, in error cases, the MEP returns Exception Codes at the Modbus response telegram. The Exception Codes are listed at the following table:

Exception Code	Name	Meaning / Possible Causes
1	Illegal Function	Unknown Function Code: Transaction contained a Modbus Function Code not supported by MEP.
2	Illegal Data Address	a) Access to unknown address (invalid xFC Function Code or Special Register) b) Error occurred while FC 43/14 transaction.
3	Illegal Data Value	a) Invalid read/write length value at Modbus transaction (Quantity and/or ByteCount) b) Malformed request telegram c) Invalid Object ID at FC 43/14 transaction.
4	Server Device Failure	Read/Write access failed (other error than unknown address)

The **Network Status LED / Sercos LED / EtherCAT Status Indicator** (Hx1) expresses the status of the MEP and the fieldbus.

For PROFINET, EtherNet/IP and ModbusTCP:

Network status LED blink code	Meaning
Continuous off	MEP does not have a valid IP address or has not seen an ethernet link, yet, or is powered off or an MEP firmware update process is running.
Blinking green	MEP does have a valid IP address, but no cyclic connection was established, yet.
Continuous green	Cyclic connection is established and it is free of errors.
Blinking red	Cyclic connection was terminated unexpectedly.
Continuous red	Duplicate IP address in network detected.
Blinking green/red	MEP is in power up mode and is conducting a self-test.

For Sercos III:

Network status LED (Sercos LED) blink code	Meaning
Continuous off	NRT state (no Sercos III communication)
Continuous orange	Communication phase 0
Orange with 1 green pulse	Communication phase 1
Orange with 2 green pulses	Communication phase 2
Orange with 3 green pulses	Communication phase 3
Continuous green	Communication phase 4

Blinking green/orange	Hotplug phase 0
Green with 1 orange pulse	Hotplug phase 1
Green with 2 orange pulses	Hotplug phase 2
Blinking green	Switched from Fast-Forward to Loopback (e.g. due to link loss at one port)
Blinking red/orange	Application error
Blinking green/red	Warning for MST losses exceeded half of tolerable losses
Continuous red	Communication error
Blinking orange	Identification
Blinking red	Firmware watchdog error

For EtherCAT:

Color Code	Meaning	Description
Off	Status INIT	- Cyclic process data and acyclic data channels are not transmitted. - No error
Blinking green	Status PRE-OPERATIONAL	Acyclic data channel is transferred
Green light blinking once	Status SAVE-OPERATIONAL	Acyclic data channel is transferred
Green light steady on	Status OPERATIONAL	Cyclic process data and acyclic data channels are transmitted
Blinking red	Configuration error	General EtherCAT configuration error
Red light blinking once	Synchronization error	- The drive controller is not synchronized to the EtherCAT master. - Communication error of the drive controller
Red light blinking twice	Timeout – watchdog	- Timeout during cyclic process data monitoring - Watchdog of the EtherCAT master

The **Module Status LED** (Hx2) expresses the status of drive application:

Module status LED blink code	Meaning
Continuous off	Drive is powered off or no communication between fieldbus platform and base drive system.
Blinking green	Drive in STOP state, no errors pending.
Continuous green	Drive in RUN state, no errors pending.
Blinking red	Drive in RUN state, a warning is pending.
Continuous red	Drive in STOP state, an error is pending.
Blinking green/red	N/A

The **PHY status LEDS** (Hx3/Hx4) are expressing the Ethernet line status:

For all fieldbusses, except EtherCAT:

PHY status LED	Meaning
Continuous off	No Ethernet link established at the appropriate Ethernet port.
Green	Ethernet link was established
Flickering yellow	Activity on Ethernet line (telegram transmission)

For EtherCAT:

PHY status LED	Meaning
Continuous off	No Ethernet link established at the appropriate Ethernet port.
Green	Ethernet link was established
Flickering green	Activity on Ethernet line (telegram transmission)

Remark: With EtherCAT, the yellow PHY LEDs are not used.

4 System Update

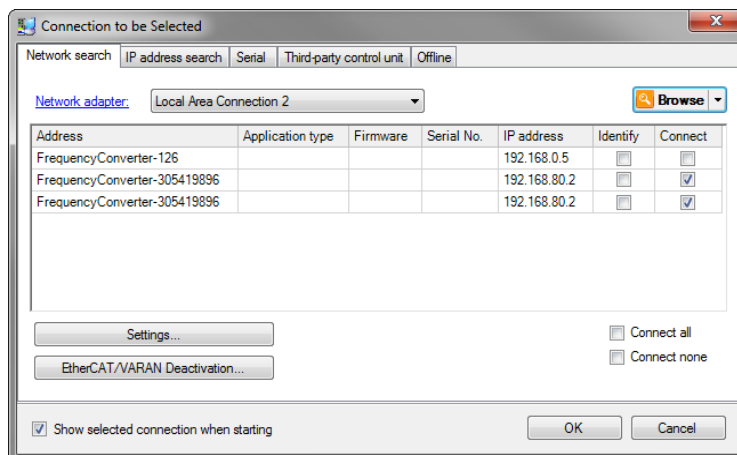
Currently, with an EFC, comprising the option “Multi-Ethernet Platform”, there are two firmware files (FWA files):

- **Base System Firmware**
Filename is of the form FWA-EFC01_-NNN-xxVyy-NN.ibf
- **Multi-Ethernet Platform Firmware**
Filename is of the form FWA-xFC01_-MEP-xxVyy-NN.ibf

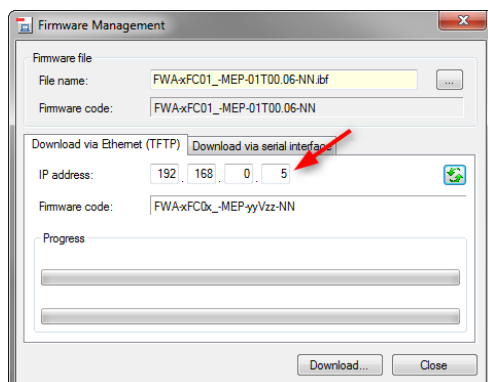
While Base System Firmware up to now can only be updated via USB port, the Multi-Ethernet Platform Firmware can only be updated via Ethernet using TFTP protocol.

Updating Multi-Ethernet Platform Firmware is rather easy done using IndraWorks. Follow these steps:

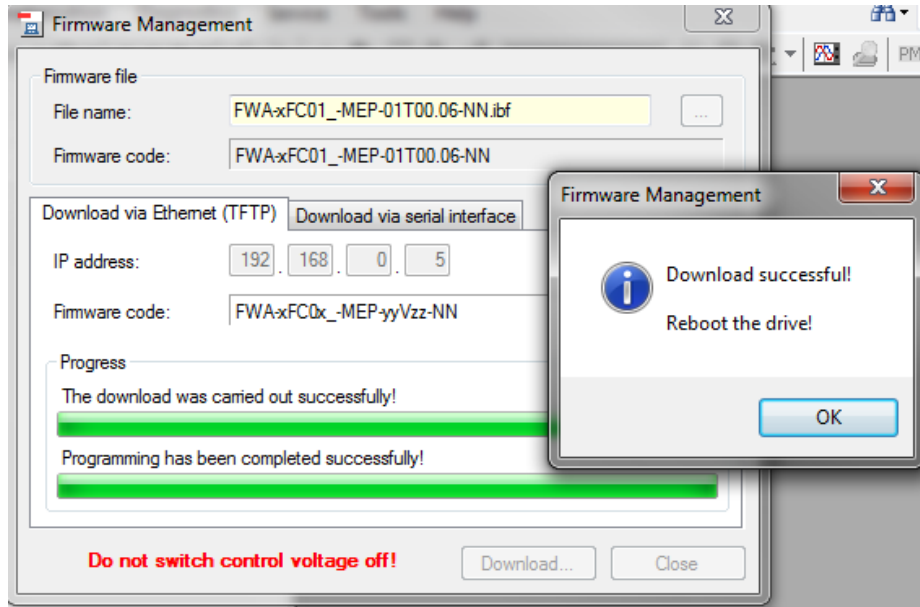
1. Connect EFC with a PC running IndraWorks via Ethernet.
2. Get current IP address of EFC to be updated using the Browsing function.
Tools → Connection → Connection Selection...
(Hint: You must not connect for performing an update - don't click OK at this dialog!)



3. Go to IndraWorks dialog Service → Firmware Management..., browse for the MEP FWA file and enter destination IP address:



4. Start update process by clicking Download... and acknowledge the appearing question.
5. Update process will take about 3 minutes. If update was not successful, check update file and IP address and try to repeat the update process. If update was successful, please power-cycle the drive.



Remarks: Current firmware version can be requested by pressing the  button. When Firmware code field shows “FWA-xFC0x_-MEP-yyVzz-NN”, there possibly is a non-consistent MEP firmware set on board and it is highly recommended to perform a MEP firmware update. Once sent to update mode, MEP can’t return to fieldbus mode, until it was rebooted.

Alternative Updating Multi-Ethernet Platform using Windows TFTP Client:

```
tftp -i 192.168.0.1 put FWA-xFC01_-MEP-01V04-NN.ibf FWA-*FC**_-MEP
```

5 Parameter Set

5.1 Overview

Function Code	Parameter Name	Data Type	Current Length	Max Length	Data after delivery	Editable
H3.00	MEP: MAC Address Device	BYTE LIST	6	6	<MANUFACTURING>	
H3.01	MEP: MAC Address Port 1	BYTE LIST	6	6	<MANUFACTURING>	
H3.02	MEP: MAC Address Port 2	BYTE LIST	6	6	<MANUFACTURING>	
H3.03	MEP: IP Address	BYTE LIST	4	4	192, 168, 0, 1	Y
H3.04	MEP: Subnet Mask	BYTE LIST	4	4	255, 255, 255, 0	Y
H3.05	MEP: Gateway Address	BYTE LIST	4	4	0, 0, 0, 0	Y
H3.06	MEP: IP Options	DWORD			0	Y
H3.07	MEP: Local Hostname (sercos/IP, EtherNet/IP)	CHAR LIST	8	100	hostname	Y
H3.08	MEP: Application Type	CHAR LIST	19	40	Frequency Converter	
H3.10	MEP: Device ID (PROFINET)	WORD			0x2802	
H3.11	MEP: Order ID	CHAR LIST	10	20	<MANUFACTURING>	
H3.12	MEP: Product Name	CHAR LIST	3	25	MEP	
H3.13	MEP: Serial Number	ULONG			<MANUFACTURING>	
H3.14	MEP: Product Code (EtherNet/IP)	WORD			36	
H3.15	MEP: Device ID (POWERLINK)	WORD			0x2802	
H3.16	MEP: Subdevice ID (POWERLINK)	DWORD			0x011F2802	
H3.18	MEP: Visual Status Indicators	ULONG			N/A	
H3.20	MEP: Station Name (PROFINET)	CHAR LIST	6	240	axis01	Y
H3.21	MEP: Station Type (PROFINET)	CHAR LIST	22	127	Rexroth-Multi-Ethernet	
H3.22	MEP: Subdevice ID (PROFINET)	DWORD			0x011F2802	
H3.23	MEP: Device Address	WORD			1	Y
H3.24	MEP: Active Device Address (Topology)	WORD			0	
H3.25	MEP: IP address is remanent (PROFINET)	BOOL				Y
H3.26	MEP: EtherCAT List of Input Process Data (Master)	WORD	0	15		
H3.27	MEP: EtherCAT List of Output Process Data (Master)	WORD	0	15		
H3.28	MEP: Input Process Data Length (Master)	USHORT			0	
H3.29	MEP: Output Process Data Length (Master)	USHORT			0	
H3.30	MEP: List of Input Process Data	WORD LIST	2	15	0x6001, 0x1002	Y
H3.31	MEP: List of Output Process Data	WORD LIST	2	15	0x6000, 0x600A	Y
H3.32	MEP: Input Process Data Length (Slave)	USHORT			4	
H3.33	MEP: Ouput Process Data Length (Slave)	USHORT			4	
H3.34	MEP: Communication Platform State	DWORD			N/A	
H3.35	MEP: Communication Diagnosis Flags	DWORD			N/A	
H3.36	MEP: ComCycle Periods [ns]	ULONG	3	3	0,0,0	
H3.37	MEP: Communication Phase	USHORT				
H3.40	MEP: Industrial Ethernet Protocol Request	CHAR LIST	2	4	PN	Y
H3.41	MEP: Industrial Ethernet Protocol Active	CHAR LIST	2	4	PN	
H3.42	MEP: Industrial Ethernet Protocol Logicware	CHAR LIST	3	4	EPL	
H3.43	MEP: TCP Debug Services Active	BOOL			1	Y
H3.49	MEP: EtherCAT State	DWORD				
H3.51	MEP: Modbus Alternative TCP Port	USHORT			0	Y
H3.52	MEP: ModBus TCP ProductCode	CHAR	10	50	0x00642880	

H3.53	MEP: ModBus TCP Vendor Name	CHAR	16	50	Bosch Rexroth AG	
H3.55	MEP: General Logbook	CHAR	0	1024		
H3.56	MEP: System Trace	CHAR	0	1024		
H3.63	MEP: List of external parameters	WORD	41	500	N/A	
H3.71	MEP: Subsystem identification parameter	CHAR LIST	0	1024	<MANUFACTURING>	
H3.96	MEP: FWA string	CHAR LIST	0	50	<MANUFACTURING>	
H4.20	MEP: EtherCAT mode (currently, only COE is supported)	CHAR	3	4	COE	Y
H4.24	MEP: EtherCAT Profilauswahl	SHORT			-128	Y

5.2 Selected Descriptions

5.2.1 [H3.06] MEP: IP Options

Bit 0 : DHCP enabled (MEP receives IP address [H3.03] from an DHCP server), other Bits unused.

5.2.2 [H3.18] MEP: Visual Status Indicators

MEP: Link/Module/Network Status									H3.18				32 Bit			
Bit	31	30	29	28	27	260	25	24	23	22	21	20	19	18	17	16
Name	unused														Link P2 Hx3	Link P1 Hx4
Dir	r	r	r	r	r	r	r	r	r	r	r	r	r	r	r	r
Init	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	Network Status Red LED Hx1				Network Status Green LED Hx1				Module Status Red LED Hx2				Module Status Green LED Hx2			
Dir	r	r	r	r	r	r	r	r	r	r	r	r	r	r	r	r
Init	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit	Name				Funktion											
31:18					UNUSED											
17	Link P2				1 = Ethernet link present 0 = No ethernet link											
16	Link P1				1 = Ethernet link present 0 = No ethernet link											
15:12	Network Status Red LED				0 = Off 1 = Blink 1 Hz 2 = Blink 2 Hz 3 = Blink 4 Hz 4 = Steady On											
11:8	Network Status Green LED															
7:4	Module Status Red LED															

3:0	Module Status Green LED	
-----	-------------------------------	--

5.2.3 [H3.34] MEP: Communication Platform State

This parameter describes the internal communication platform state.

Parameter Value	State	Description
0	NOP	Communication platform inactive
1	START	Running boot process
3	SYSRDY	System up, preparing for configuration
4	CONFIG	System basic configuration done
6	COMCFG	Fieldbus selection done
8	COMINIT	Ready for being connected by fieldbus master
10	COMRDY	Preparing for cyclic communication
11	COMACTV	Cyclic communication active
2	STARTERR	Error at boot process
5	CFGERR	Error at system basic configuration
7	COMCFGERR	Error at fieldbus selection
9	COMINITERR	Error at fieldbus configuration
12	COMERR	Error / Breakdown of cyclic communication
13	UPDATE	Update in progress

5.2.4 [H3.35] MEP: Communication Diagnosis Flags

This parameter gives some detailed diagnosis on internal events. However, all diagnosis flags are assigned to some error codes and corresponding display messages.

Bit	Bezeichnung	Beschreibung
0	LINK Error	No ethernet link
1	IP Address Error	IP address already present at subnet
2	IP Init Error	Error while starting IP stack
3	MAC Address Error	Invalid MAC address
4	DHCP Error	DHCP request: No response from DHCP server.
5	Identify Error	Invalid identification parameters (e.g. PROFINET device name)
8	Fieldbus Init Error	Error while starting fieldbus stack
9	Connection Error	An existing cyclic communication got broken because of an communication problem.
10	Connection Idle	Fieldbus master set process data status to "invalid" (e.g. Ethernet/IP „Run/Idle-Flag changed", PROFINET: Module Status and/or APDU Status)
11	Connection closed	An existing cyclic communication was closed by the fieldbus master.
12	Connection timeout	An existing cyclic communication was terminated because of missing master telegrams.
13	PDC Difference	Process Data Configuration of Communication Platform ([H3.30]/[H3.31]) and Process Data Configuration of Fieldbus Master are differing in data length. Slave PDC length can be read from [H3.32]/[H3.33]) and Master PDC length can be read from [H3.28]/[H3.29])
14	PDC Invalid	Process Data Configuration contains unknown/unsupported parameters or exceeds maximum data length of 15 parameters for input and output data, each.
17	FW CRC-OK	Communication Platform Firmware consistency check done and ok.

26	FW CRC-Error	Communication Platform Firmware consistency check failed.
27	Host Watchdog	Internal communication to base system timed out. Timeout depends on [H3.38].
29	Parameterization error	MEP has invalid parameters (see [H3.62])

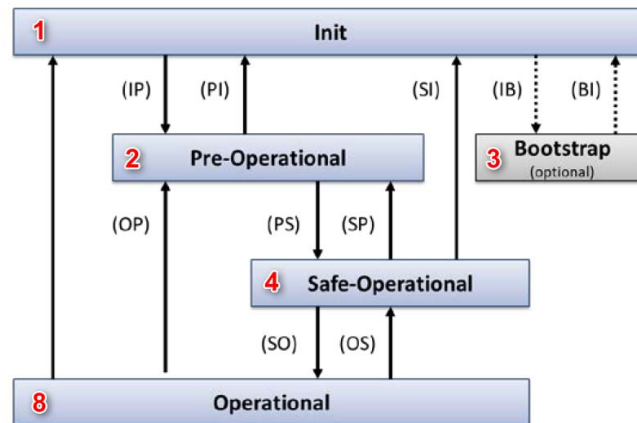
5.2.5 [H3.36] MEP: ComCycle Periods

This parameter consists of three 4-byte-values that specify current communication cycle periods. All values are in nanoseconds.

- **Value No. 1:** Transmission Cycle On Bus
- **Value No. 2:** Producer Cycle (Input Data Cycle)
- **Value No. 3:** Consumer Cycle (Output Data Cycle)

5.2.6 [H3.49] MEP: EtherCAT State

This parameter contains an encoding of the current EtherCAT state, according to following assignment:



6 Reporting Problems

This section gives some assistance in documenting and reporting about unexpected behavior of the Multi-Ethernet Platform. Because of many possible influences on fieldbus problems, it is important to submit sufficient information about an occurred problem.

- **Network Topology**
Document all network participants, their node-specific information (device type, MAC/IP address, PNIO Station Name) and their connectivity inside the network.
- **Hardware / Firmware Version**
Document the used hardware and firmware version. For Multi-Ethernet Platform all this information is contained at [H3.71]. (Note, that this information is already contained in a parameter file drawn from the EFC – see below.)
- **Error Causing Actions**
Document the status of all components previous to the error, and the (user/PLC) actions, which led to the error (e.g. plugged/pulled ethernet cable, hardware exchanged etc.)
- **Error Description**
Describe the *expected* reaction of involved devices. Describe the *observed* reaction of involved devices. Describe from your view, why the observed reaction must be erroneous.
- **Parameter Settings**
Generate parameter files of all devices involved with the problem *after* the problem occurred and submit the files.
- **Recording Ethernet Traffic**
In case of fieldbus traffic problems, try to record the ethernet traffic around occurrence of the error. (Use Wireshark/Ethereal/Packetyzer in conjunction with a monitor-ported switch or use some ethernet taps.)

Your feedback is highly appreciated – thank you for your patience!

7 Fallacies and Pitfalls

This section contains typical problems occurred with Multi-Ethernet Platform reported by users.

7.1 Sercos III SDDML

Observation: When connecting to EFC running Sercos, the Vendor Device Id (S-0-1300.0.5) does not match.

Cause: SDDML file contains Vendor Device Id “EFC01” (approx. at line 219). However, there are some older EFCs, whose Device Hardware Label [b8.00] contains a “vid” value of “xFC0x”.

Workaround: Change SDDML file for those old EFCs to Vendor Device Id “xFC0x”.

7.2 Error “Fin-“ after updating MEP firmware

Observation: After performing an update to a new MEP firmware version and reboot, MEP signals error “Fin-“ at the LED panel.

Cause: After performing an update to a new MEP firmware version, there could be some new parameters, which are not set valid, yet.

Solution: To clear the error, look at [H3.62] and write all parameters listed once with default value or perform a parameter base load process, which includes also the communication parameter set.

Example: Three parameters are invalid, when updating from MEP 01V02 to 01V04:

